

Checkpoint-RelAlg Formula Source

F01 — Rendered causal context.

$$M_t = \text{Render}(Q, K, T_t, C_{0:k}, S_t, E_t).$$

F02 — Exactly one authored action.

$$a_t \sim \pi_\theta(\cdot \mid M_t), \quad |\text{AuthoredActions}_t| = 1.$$

F03 — Authority partition.

$$\mathcal{C}_{\text{task}} = Q \oplus K, \quad \mathcal{F}_{\text{db}} = S_t, \quad C_{0:k} \not\models \text{database facts}.$$

F04 — Mode tool surfaces.

$$\mathcal{T}_D = \mathcal{P} \cup \{\text{execute_sql}\} \cup \mathcal{C}, \quad \mathcal{T}_A = \mathcal{P} \cup \mathcal{R} \cup \mathcal{C}, \quad \mathcal{T}_H = \mathcal{T}_D \cup \mathcal{T}_A.$$

F05 — Immutable relation artifact.

$$A = \langle h, k, \langle c_i : \tau_i \rangle_{i=1}^m, n, \omega, d \rangle.$$

F06 — Selected relational operators.

$$\sigma_p(R), \quad \pi_{\langle e_i \rightarrow a_i \rangle_{i=1}^m}(R), \quad R \bowtie_\theta S, \quad \gamma_{G; \langle f_i(c_i) \rightarrow a_i \rangle}(R), \quad \tau_K(R).$$

F07 — Bag join multiplicity.

$$\mu_{R \bowtie_\theta S}(r, s) = \mu_R(r) \mu_S(s) \mathbf{1}[\theta(r, s)].$$

F08 — Checkpoint snapshot.

$$C_k = \langle D_k, A_k, O_k, U_k, h_k \rangle, \quad h_k = \text{SHA256}(\text{canon}(S_k)).$$

F09 — Restore transition.

$$\text{Restore}(C_j) : \quad S \leftarrow S(C_j), \quad \mathcal{P}_{\text{active}} \leftarrow \text{Ancestors}(C_j) \cup \{C_{\text{recovery}}\}.$$

F10 — State-preserving failure.

$$S_{t+1} = \begin{cases} \text{Apply}(S_t, a_t), & \text{valid}(a_t), \\ S_t, & \text{rejected}(a_t) \vee \text{failed}(a_t), \end{cases} \quad H(S_{t+1}) = H(S_t) \text{ on failure}.$$

F11 — Grounded answer and official score.

$$\widehat{R} = \text{Artifact}(a_T), \quad \text{Acc}_{\text{official}} = \mathbf{1}[\text{Set}(\widehat{R}) = \text{Set}(R^*)].$$

F12 — Strict artifact audit.

$$\text{Strict}(\widehat{R}, R^*) = \text{Schema} \wedge \text{Rows}_{\text{multiset/sequence}} \wedge \text{NULL} \wedge \text{Multiplicity}.$$